

# Guideline

## Adult head injuries in the emergency department

### 1 Scope

Patients in the emergency department (ED).

### 2 Purpose

To help the clinician assessing adult patients (over 16's) with head injuries.

### 3 Introduction

The following advice is based on Advanced Trauma and Life Support (ATLS) principles and a policy that has evolved and been agreed between the neurosurgical, radiology and emergency departments.

### 4 Guideline for managing head injuries

Head injuries are classified as **mild** (Glasgow coma scale (GCS) 13-15), **moderate** (GCS 9-12) and **severe** GCS (3-8) depending on the initial Glasgow coma scale (GCS) (after primary survey resuscitation).

#### 4.1 Moderate (GCS 9-12) and severe GCS (3-8)

All patients with GCS less than 13 should be assessed within the resuscitation room of ED. They are likely to require activation of the trauma team and will be treated according to ATLS methods of resuscitation, paying attention to the airway, breathing and circulation (ABCs). A mini neurological assessment must be carried out and recorded. This will consist of the GCS, examination of the pupils and any gross neurological deficit. All such patients will require urgent computed tomography (CT) of the head and neck and referral to the neurosurgical department. Such patients are admitted under the neurosurgical team.

Appropriate attention and management will also be given to other injuries present as part of major trauma management (MTC). Neuro-protective measures should be initiated to all ventilated HI patients.

Neuroprotective measures:

- Avoidance of hypotension – target BP >100mmHg in 50-69 year olds, target BP >110mmHg in 15-49 years old and over 70 years old.
- Maintain normal oxygenation (normal PaO<sub>2</sub>)
- Target PaCO<sub>2</sub> 4.5-5.0 KPa
- Avoidance of fever
- Target normoglycaemia
- Tilt trolley to achieve 30 degrees head up elevation

## Emergency department

### Emergency and perioperative care directorate

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- Aim to release cervical collar early (once the c-spine is cleared of injury)
- Tape rather than tie endotracheal tubes in place
- Prevent and aggressively manage seizures

#### 4.2 Mild head injuries (GCS 13-15)

A thorough assessment of these patients is carried out particularly noting any of the following features:

- Witnessed loss of consciousness or amnesia.
- Mechanism of injury.
- Time of injury
- Initial GCS
- GCS in the Emergency Department & any trend up or down
- Any abnormality on neurological examination.
- Seizures
- Headache severity
- Suspected penetrating injury, depressed skull or signs of base of skull fracture.
- Elderly patients: the use of anticoagulants or the presence of bleeding diathesis, any other significant medical conditions.
- Difficulty in assessing the patient (eg alcohol, epilepsy, inadequate or suspicious history).
- Neurological symptoms, especially vomiting more than once or the presence of neurological deficit.

#### 4.3 Management

Note that there are different care pathways for children and adults/ adolescent children. These are covered in the document “Head injury in children” at <http://merlin/Lists/DMSRecords/DispRecordTabsDoc.aspx?ID=18034&IsDlg=1&Source=/&IsDlg=1#>

Management of Adults:

- Patients with no indications for a head CT can be discharged with written advice (see 4.8 below).
- Appropriate patients can be admitted to the ED Observation Ward (Clinical Decisions Unit) under the care of ED consultants.
- If no bed is available on the Observation Ward then the patient should be admitted to an inpatient bed on a ward capable of carrying out head injury observations, usually under the care of either Neurosurgery or Medicine for the Elderly. Choice of the Admitting Speciality will depend on the

## Emergency department

### Emergency and perioperative care directorate

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patient's dominant injury/problem requiring admission, as well as any comorbidities & social issues preventing discharge.

- Patients whose primary problem requiring admission is their head injury should be admitted under neurosurgery.
- Patients with a head injury not requiring Neurosurgical intervention **and** complex comorbidities or decompensation of existing medical problems may be more appropriate for admission under Medicine or Care of the Elderly. If in doubt discuss with an ED Consultant prior to referral.
- The remaining patients will be discharged home from the department.

#### 4.4 Criteria for CT scanning

Adults will be investigated according to the NICE Guideline No. 176.

4.4.1 The following head injured adult patients should undergo CT scanning within 1 hour of the risk factor being identified:

- GCS less than 13 on initial assessment in the emergency department.
- GCS less than 15 at two hours after the injury on assessment in the emergency department.
- Suspected open or depressed skull fracture.
- Any sign of basal skull fracture (haemotympanum, 'panda' eyes, cerebrospinal fluid leakage from the ear or nose, Battle's sign).
- Post-traumatic seizure.
- Focal neurological deficit
- More than one episode of vomiting.

Adults with any of the risk factors below who have experienced amnesia or loss of consciousness since the injury, should have a CT head performed within 8 hours of injury:

- Age 65 or over
- Any history of bleeding or clotting disorders
- Dangerous mechanism of injury (pedestrian/cyclist hit by a car, ejected from a motor vehicle or a fall from greater than 1 metre or 5 steps).
- More than 30 minutes of retrograde amnesia of events immediately before the head injury.

Adults who have sustained a head injury with no further indication for a CT head scan and who are on **anticoagulant treatment** require a head CT. Perform this within eight hours of the injury.

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#### 4.5 Cervical spine CT scanning

For adults who have sustained a head injury and have any of the following risk factors, perform a CT cervical spine scan within one hour of the risk factor being identified:

- GCS less than 13 on initial assessment.
- The patient has been intubated.
- Plain x-rays are technically inadequate (for example, the desired view is unavailable).
- Plain x-rays are suspicious or definitely abnormal.
- A definitive diagnosis of cervical spine injury is needed urgently (for example, before surgery).
- The patient is having other body areas scanned for head injury or multi-region trauma.
- The patient is alert and stable, there is clinical suspicion of cervical spine injury and any of the following apply:
  - Age 65 years or older
  - Dangerous mechanism of injury (fall from a height greater than one metre or five stairs; axial load to the head, for example, diving; high-speed motor vehicle collision; rollover motor accident; ejection from a motor vehicle; accident involving motorised recreational vehicles; bicycle collision)
  - Focal peripheral neurological deficit
  - Paraesthesia in the upper or lower limbs.

#### 4.6 Criteria for neurosurgical referral

- Coma continuing after resuscitation - even if no skull fracture.
- Deterioration in level of consciousness or progressive neurological deficit.
- Open injury - depressed fracture or penetrating injury.
- Abnormal CT.
- Patients with a GCS≤12 after resuscitation.

#### 4.7 Criteria for admission to the observation ward or the hospital

- Confusion or reduced level of consciousness at the time of examination.
- Simple skull fracture (with normal CT scan) or clinical signs of basal skull fracture.
- Difficulty in assessing the patient eg suspected drugs, alcohol, non-accidental injury, epilepsy, attempted suicide.

## Emergency department

### Emergency and perioperative care directorate

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- Other medical conditions (eg coagulation disorders).
- Patient's social circumstances or lack of a responsible adult at home to observe them post head injury.

Post-traumatic amnesia (<2 minutes) or transient unconsciousness with full recovery are not necessarily indications for admission, (if in doubt discuss and/or admit).

Patients in the above groups are suitable for admission to the observation ward under the care of the ED consultant.

On occasions patients may remain on the ward under the ED consultants, even if abnormalities are demonstrated on CT. However, the neurosurgical registrar should be asked for a review for any CT thought to be abnormal. If no beds are available in CDU, the patient must be referred to neurosurgery for admission to an in-hospital bed.

#### 4.8 Discharge arrangements

All of the following criteria should be fulfilled: ·

GCS 15

No criteria for observation

No indication for CT / normal CT

Verbal and written head injury advice given with explanation of the "red flags" that should trigger return to the ED.

GP letter completed ·

A responsible adult must be available to accompany the patient home.

The ED patient advice leaflet is available at

<http://merlin/Lists/DMSRecords/DispRecordTabsDoc.aspx?ID=18652&IsDlg=1&Source=/&IsDlg=1#>

Alternative patient discharge advice is available from NICE at:

<https://www.nice.org.uk/guidance/cg176/resources/head-injury-suggested-written-discharge-advice-pdf-191717245>

A neurosurgical head injury clinic is available for follow up 2-4 weeks post injury. This clinic should be reserved for those with ongoing significant symptoms or CT abnormalities e.g. in patients with minor cerebral contusions on CT or significant post-concussion symptoms. Discuss the need for this with the Neurosurgical registrar or consultant.

Patients with facial fractures are followed up in the maxillo-facial department.

#### 4.9 Post-concussion syndrome- PCS

Some patients experience unpleasant symptoms for days or several weeks after a head injury, may not be aware of this and may become very concerned, anxious, and sometimes depressed. They may be afraid they have 'brain

## Emergency department

### Emergency and perioperative care directorate

damage' or that they will never be back to normal and may make wrong decisions about their future. They will suffer from a variety of symptoms which include headaches, drowsiness, lack of memory and concentration, tiredness, dizziness, irritability, clumsiness and problems with schooling. You must exclude serious intracranial pathology with a head CT.

It is important to warn patients and their relatives about such symptoms and to reassure them that they will improve and return to normality.

An information leaflet on Concussion is available on CONNECT at <http://merlin/Lists/DMSRecords/DispRecordTabsDoc.aspx?ID=18711&ContentTypeld=0x01004652AD2286CC764AB5A9F43F02C29047003288753F9176604788ABF0A752F4C786#>

Paper copies are kept in the ED Minors and on CDU.

Written discharge advice should be given to the patient and their relatives. It has been found to be effective in aiding smooth recovery. Initial rest and simple analgesics are helpful, followed by gradual return to normal activities. Sporting activities should be avoided until full recovery, as patients with PCS are susceptible to further HI and concussion.

If patients continue to have symptoms 4-6 weeks post-head injury, they should visit their GP for referral to the Addenbrookes HI clinic. Alternatively, if the patient's PCS symptoms are severe at first presentation and you anticipate a long recovery, you can refer them to the neurosurgical head injury clinic (via EPIC) as mentioned above. If in doubt, please consult with a senior.

## 5 Monitoring compliance with and the effectiveness of the clinical guideline

Clinical breaches of this standard will be raised by the incident reporting system.

## Equality and diversity statement

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## Document management

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